

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: CYBER & INFORMATION SECURITY

Practical No. 5

AIM: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

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CODE:

```
T098 Vikas Cyber Pract 5.py - C:/Users/PRIYANKA/Downloads/T098 Vikas Cyber Pract 5.py (3.13.1)
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# Diffie-Hellman Key Exchange

print("===== DIFFIE-HELLMAN KEY EXCHANGE =====")

# Public values
p = 23
g = 5

print("\nPublic values:")
print("Prime (p):", p)
print("Generator (g):", g)

# Private keys
a = int(input("\nEnter private key of Alice: "))
b = int(input("Enter private key of Bob: "))

# Calculate public keys
A = pow(g, a, p)
B = pow(g, b, p)

print("\nAlice's public key:", A)
print("Bob's public key:", B)

# Calculate shared secret
alice_secret = pow(B, a, p)
bob_secret = pow(A, b, p)

print("\nAlice's shared secret:", alice_secret)
print("Bob's shared secret:", bob_secret)

# Verify
if alice_secret == bob_secret:
    print("\n✓ Key exchange successful!")
    print("Shared secret key:", alice_secret)
else:
    print("\n✗ Key exchange failed!")
|
```

OUTPUT:

```
Windows PowerShell
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PS C:\Users\PRIYANKA> python "C:\Users\PRIYANKA\Downloads\T098 Vikas Cyber Pract 5.py"
===== DIFFIE-HELLMAN KEY EXCHANGE =====

Public values:
Prime (p): 23
Generator (g): 5

Enter private key of Alice: 56
Enter private key of Bob: 98

Alice's public key: 18
Bob's public key: 9

Alice's shared secret: 9
Bob's shared secret: 9

✓ Key exchange successful!
Shared secret key: 9
PS C:\Users\PRIYANKA>
```

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CODE GUI:

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import tkinter as tk
from tkinter import messagebox

# ----- DIFFIE-HELLMAN VALUES -----
p = 23
g = 5

# ----- FUNCTIONS -----
def generate_key():
    try:
        a = int(alice_entry.get())
        b = int(bob_entry.get())

        if a <= 0 or b <= 0:
            messagebox.showerror("Error", "Enter positive private keys.")
            return

        # Public keys
        A = pow(g, a, p)
        B = pow(g, b, p)

        # Shared secret keys
        alice_key = pow(B, a, p)
        bob_key = pow(A, b, p)

        # Display results
        alice_public.config(text=str(A))
        bob_public.config(text=str(B))

        alice_shared.config(text=str(alice_key))
        bob_shared.config(text=str(bob_key))

        if alice_key == bob_key:
            result.config(
                text="✓ Key Exchange Successful",
                fg="green"
            )
            result_sub.config(
                text="Both Alice and Bob have the same shared secret key.",
                fg="green"
            )

    except ValueError:
        messagebox.showerror(
            "Error",
            "Please enter valid numbers."
        )

def clear():
    alice_entry.delete(0, tk.END)
    bob_entry.delete(0, tk.END)

    alice_public.config(text="-")
    bob_public.config(text="-")
    alice_shared.config(text="-")
    bob_shared.config(text="-")

    result.config(
        text="Result will appear here",
        fg="black"
    )
    result_sub.config(
        text="",
        fg="black"
    )

# ----- MAIN WINDOW -----
root = tk.Tk()

root.title("Diffie-Hellman Key Exchange")
root.geometry("850x650")
root.resizable(False, False)
root.configure(bg="white")

# ----- TITLE -----
title = tk.Label(
    root,
    text="Diffie-Hellman Key Exchange",
    font=("Arial", 25, "bold"),
    bg="white"
)
```

```
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except ValueError:
    messagebox.showerror(
        "Error",
        "Please enter valid numbers."
    )

def clear():
    alice_entry.delete(0, tk.END)
    bob_entry.delete(0, tk.END)

    alice_public.config(text="-")
    bob_public.config(text="-")
    alice_shared.config(text="-")
    bob_shared.config(text="-")

    result.config(
        text="Result will appear here",
        fg="black"
    )
    result_sub.config(
        text="",
        fg="black"
    )

# ----- MAIN WINDOW -----
root = tk.Tk()

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fg="#17388a"
)
title.pack(pady=(25, 20))

# ----- PUBLIC VALUES -----

public_frame = tk.LabelFrame(
    root,
    text="Public Values",
    font=("Arial", 13, "bold"),
    bg="white",
    fg="#17388a",
    padx=20,
    pady=15
)

public_frame.pack(
    fill="x",
    padx=35
)

tk.Label(
    public_frame,
    text="Prime (p):",
    font=("Arial", 11),
    bg="white"
).grid(row=0, column=0, padx=(180, 10), pady=5)

p_value = tk.Entry(
    public_frame,
    width=10,
    font=("Arial", 11),
    justify="center"
)

p_value.insert(0, "23")
p_value.config(state="readonly")

p_value.grid(row=0, column=1, padx=10)

tk.Label(
    public_frame
```

```
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    padx=20,
    pady=20
)

alice_frame.grid(
    row=0,
    column=0,
    sticky="nsew",
    padx=(0, 10)
)

tk.Label(
    alice_frame,
    text="Enter Private Key (a):",
    font=("Arial", 11),
    bg="white"
).grid(row=0, column=0, padx=10, pady=10)

alice_entry = tk.Entry(
    alice_frame,
    width=20,
    font=("Arial", 11)
)

alice_entry.grid(
    row=1,
    column=0,
    padx=10
)

# Bob

bob_frame = tk.LabelFrame(
    users_frame,
    text="Bob (User B)",
    font=("Arial", 13, "bold"),
    bg="white",
    fg="green",
    padx=20,
    pady=20
)

bob_frame.grid(
```

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row=0,
column=1,
sticky="nsew",
padx=(10, 0)
)

tk.Label(
    bob_frame,
    text="Enter Private Key (b):",
    font=("Arial", 11),
    bg="white"
).grid(row=0, column=0, padx=10, pady=10)

bob_entry = tk.Entry(
    bob_frame,
    width=20,
    font=("Arial", 11)
)

bob_entry.grid(
    row=1,
    column=0,
    padx=10
)

users_frame.columnconfigure(0, weight=1)
users_frame.columnconfigure(1, weight=1)

# ----- BUTTONS -----

button_frame = tk.Frame(
    root,
    bg="white"
)

button_frame.pack(pady=5)

tk.Button(
    button_frame,
    text="Generate Shared Key",
    font=("Arial", 11)

```

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    padx=10
)

tk.Button(
    button_frame,
    text="Clear",
    font=("Arial", 11),
    width=12,
    command=clear
).grid(
    row=0,
    column=1,
    padx=10
)

# ----- RESULTS -----

result_frame = tk.LabelFrame(
    root,
    text="Results",
    font=("Arial", 13, "bold"),
    bg="white",
    fg="#17388a",
    padx=30,
    pady=20
)

result_frame.pack(
    fill="x",
    padx=35,
    pady=20
)

# Alice Public

tk.Label(
    result_frame,
    text="Alice Public Key (A):",
    font=("Arial", 11),
    bg="white"
).grid(row=0, column=0, padx=20, pady=8)
```

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```
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text="Alice Public Key (A):",
font=("Arial", 11),
bg="white"
).grid(row=0, column=0, padx=20, pady=8)

alice_public = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11),
    bg="white",
    width=12,
    relief="solid"
)

alice_public.grid(row=0, column=1, padx=20)

# Bob Public

tk.Label(
    result_frame,
    text="Bob Public Key (B):",
    font=("Arial", 11),
    bg="white"
).grid(row=0, column=2, padx=20)

bob_public = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11),
    bg="white",
    width=12,
    relief="solid"
)

bob_public.grid(row=0, column=3, padx=20)

# Alice Shared

tk.Label(
    result_frame,
    text="Alice Shared Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=0, padx=20)

# Bob Shared

tk.Label(
    result_frame,
    text="Bob Shared Key:",
    font=("Arial", 11),
    bg="white"
).grid(row=1, column=2, padx=20)

bob_shared = tk.Label(
    result_frame,
    text="-",
    font=("Arial", 11),
    bg="white",
    width=12,
    relief="solid"
)

bob_shared.grid(row=1, column=3, padx=20)

# ----- SUCCESS MESSAGE -----

success_frame = tk.Frame(
    root,
    bg="#e0f0e0",
    highlightbackground="#b7d7a8",
    highlightthickness=1
)

success_frame.pack(
    fill="x",
    padx=35,
    pady=(0, 15)
)

result = tk.Label(
    success_frame,
    text="Result will appear here",
    font=("Arial", 15, "bold")
)
```

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bob_shared.grid(row=1, column=3, padx=20)

# ----- SUCCESS MESSAGE -----

success_frame = tk.Frame(
    root,
    bg="#e0f0e0",
    highlightbackground="#b7d7a8",
    highlightthickness=1
)

success_frame.pack(
    fill="x",
    padx=35,
    pady=(0, 15)
)

result = tk.Label(
    success_frame,
    text="Result will appear here",
    font=("Arial", 15, "bold"),
    bg="#e0f0e0",
    fg="black"
)

result.pack(pady=(12, 2))

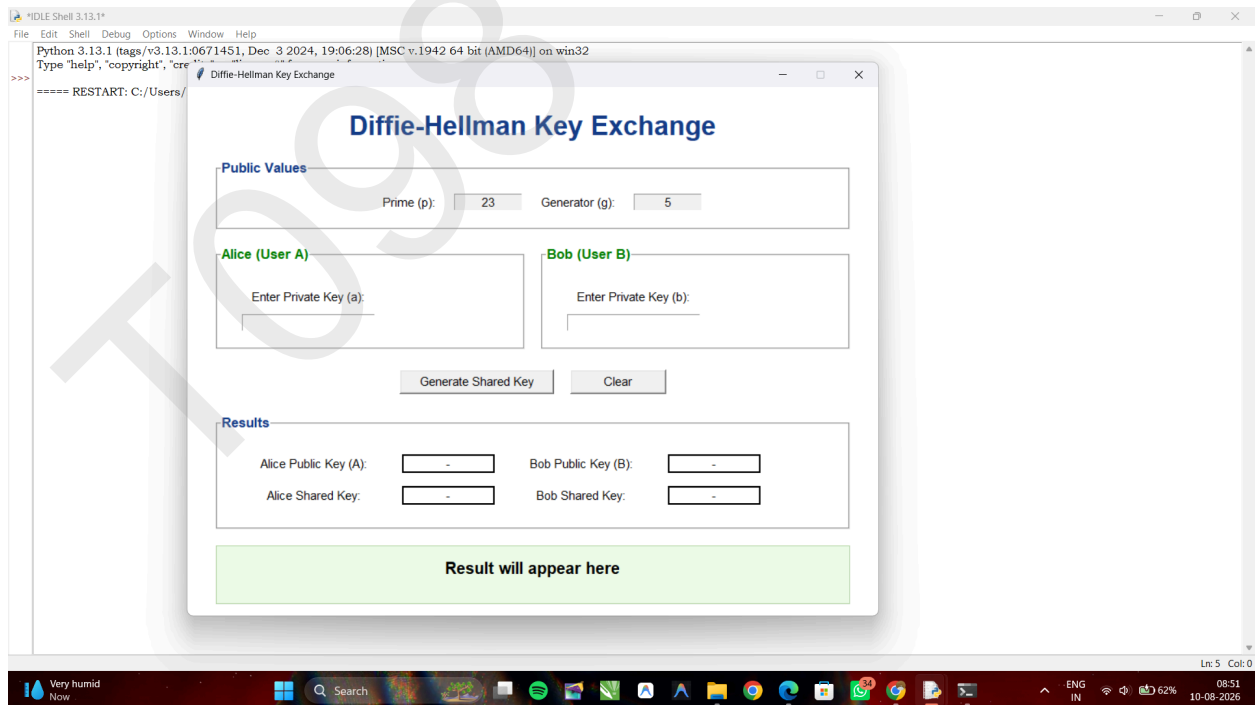
result_sub = tk.Label(
    success_frame,
    text="",
    font=("Arial", 10),
    bg="#e0f0e0"
)

result_sub.pack(pady=(0, 12))

# ----- RUN -----

root.mainloop()
```

OUTPUT:



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